Acta Mechanica Solida Sinica
https://doi.org/10.1007/s10338-026-00792-z

Incremental Elasticity of Pre-stretched Elastomers with Strain Hardening Effect: Half-Space and Thin-Layer Limits

Le Du[1] · Zhaohe Dai[2] · Jianmin Long[3] · Rui Xiao[1] · Weiqiu Chen[1]

Received: 25 February 2026 / Revised: 16 May 2026 / Accepted: 18 May 2026

Abstract

Soft materials, such as rubber, biological tissues, and hydrogels, typically exhibit strain hardening effect that preserves their structural integrity and function. Moreover, these materials are rarely stress- or strain-free in their natural or engineered states. The presence of pre-stress and pre-strain significantly influences their mechanical properties. In this paper, we investigate the mechanical response of pre-stretched, strain hardening soft elastomers in two limiting cases: the half-space and the thin layer. By employing the Fourier transform, we derive the surface Green's function for the half-space. A perturbation approach is utilized to obtain an analytical expression for the surface deformation of a pre-stretched thin layer bonded to a rigid substrate. Based on these derivations, spherical indentation problems for both limiting configurations are analyzed. The relationship between indentation force and indentation displacement is influenced by the pre-stretch state and the number of chain segments of the elastomer, i.e., the hardening parameter. The theoretical results are consistent with finite element simulations. Overall, this study provides a versatile theoretical basis for characterizing the mechanical properties and analyzing other surface phenomena of pre-stretched soft materials with strain hardening effect.

Keywords Pre-stretch · Strain hardening · Half-space · Thin layer · Incremental elasticity

1 Introduction

Soft materials, such as elastomers, biological tissues, and hydrogels, are ubiquitous in biological systems and widely utilized in biomedical engineering. These materials rarely exist in a stress-free or strain-free state; rather, the presence of pre-stress and pre-strain significantly affects their mechanical properties and functional performance [1–5]. For example, in different biological scenarios, such internal states may be either detrimental [6] or beneficial [7]. Additionally, recent single-asperity contact experiments using rigid diamond tips have demonstrated that the frictional response of polymer layers is strongly dependent on the degree of pre-stretch [8]. In practical applications, pre-stressed or pre-strained soft materials inevitably undergo surface interactions with other objects. Further, indentation tests based on incremental elasticity have emerged as effective tools for characterizing pre-stresses in materials, even their degree of anisotropy [9–11]. Therefore, investigating the incremental elasticity of pre-stressed and pre-strained soft materials is of fundamental importance.

Previous studies have made valuable contributions to the understanding of pre-stretched soft materials. On the one hand, based on the surface Green's function of an incompressible neo-Hookean elastomer subject to biaxial pre-stretching developed by He [12], Zheng et al. [9] investigated the contact between rigid cylindrical and spherical indenters and non-equal biaxially stretched substrates, revealing the significant influence of the pre-stress state on indentation behavior. Utilizing the same surface Green's function, Zheng and Cai [10], as well as Du et al. [13], explored contact problems of flat-ended elliptical, ellipsoidal, and elliptic conical indenters with pre-stretched elastomers, accounting for the relative angle between the indenter and the principal stretching directions. The results indicated that the shape of the contact region is governed by both the pre-stretch state and the indenter geometry. Yuan et al. [14] and Liang et al.

✉ Rui Xiao
rxiao@zju.edu.cn

1 Department of Engineering Mechanics, Zhejiang University, Hangzhou 310027, China

2 School of Mechanics and Engineering Science, Peking University, Beijing 100871, China

3 College of Mechanics and Engineering Science, Hohai University, Nanjing 210098, China

Published online: 09 July 2026

[15] employed finite element methods (FEMs) to study the effects of equibiaxial residual stress on indentation for an ideal elasto-plastic model and one with post-yield hardening, respectively. On the other hand, by incorporating finite thickness into the surface Green's function, Song et al. [16] and Lv et al. [17] investigated the indentation of flat-ended cylindrical and square indenters on finite-thickness layers bonded to rigid substrates, respectively, showing that indentation force increases markedly as thickness decreases. The presence of pre-stress also substantially affects adhesion behavior. Pioneering work by He and Ding [18] addressed the adhesion between an equibiaxially pre-stretched elastomer and a rigid sphere. Subsequently, Frétigny and Chateauminois [19] and Barney and Zheng [20] theoretically investigated the adhesion between a rigid spherical indenter and a non-equibiaxially pre-stretched substrate using different assumptions for the pressure distribution. Recently, the incremental elasticity of pre-deformed compressible soft magnetoactive materials has been extensively investigated [21–23], with particular consideration given to the electromechanical coupling effect, thereby advancing the application of such materials in soft robotics and smart actuators.

However, the aforementioned studies primarily employed neo-Hookean or elasto-plastic models, which cannot describe the strain hardening effect of soft materials under finite deformation. In reality, many soft materials, including rubber and biological gels, exhibit pronounced stiffening or hardening at large strains [24–27]. The strain hardening behavior provides compliance to small forces while offering resistance against large, potentially damaging deformations, thereby maintaining structural integrity and function [28–30]. Consequently, an incremental elasticity theory that accounts for the strain hardening effect in pre-stretched elastomers is essential. However, this problem remains largely unsolved. To describe strain hardening induced by finite chain extensibility, constitutive models such as the Gent [31] and Arruda-Boyce models [32] are commonly used. These models have been shown to provide nearly equivalent predictive capabilities with only two parameters [33]. The Gent model provides a phenomenological description by introducing a limiting value of the first strain invariant, whereas the Arruda-Boyce model is based on molecular chain statistics and relates the hardening parameter to the number of chain segments. Liang et al. [34] provided an analytical surface Green's function for an equibiaxially stretched half-space based on the Gent model [31] and investigated flat-ended cylindrical indentation. Nevertheless, the indentation response of strain hardening pre-stretched elastomers to other indenter geometries remains to be investigated. Moreover, thin-film structures are prevalent in numerous biological and engineering contexts [35–39]. For example, thin elastomeric or gel coatings are used to enhance the biocompatibility of medical devices [40, 41]. In these scenarios, where the thickness of the thin films is small relative to the lateral dimensions, half-space solutions become inapplicable [42, 43].

This paper aims to investigate the incremental elasticity of pre-stretched elastomers exhibiting strain hardening effects based on the Arruda-Boyce model [32], focusing on two limits: the half-space and the thin layer. The remainder of this paper is organized as follows: In Sect. 2, we derive the surface Green's function for a half-space using the Fourier transform and employ a perturbation approach to obtain the surface deformation for a pre-stretched thin layer bonded to a rigid substrate. In Sect. 3, we discuss the influences of pre-stretch and strain hardening parameter on indentation behavior through two case studies: rigid spherical indentation on a pre-stretched half-space and on a pre-stretched thin layer. Section 4 provides a summary of our findings.

2 Problem Description and General Formulation

2.1 Surface Green's Function in the Half-Space Limit

We begin with the case of the half-space limit. Consider an incompressible, strain hardening elastomer described by the Arruda-Boyce hyperelastic model, occupying the infinite half-space $x_3 \leq 0$ in a Cartesian coordinate system $Ox_1x_2x_3$. The initial homogeneous and isotropic state is defined as the reference configuration C_0, as shown in Fig. 1a. The elastomer undergoes a sequential two-step deformation: first, it is subjected to finite biaxial stretches λ_1 and λ_2 along the x_1- and x_2-directions, respectively, reaching the intermediate configuration C_1, as presented in Fig. 1b. Subsequently, a concentrated force $f_i (i = 1, 2, 3)$ is applied to the surface $x_3 = 0$, inducing infinitesimal incremental displacements u_i and leading to the current configuration C_2, as shown in Fig. 1c. Let X_A, $\bar{x}_i$, and x_i denote the coordinates of a representative material point in C_0, C_1, and C_2, respectively. Herein, an overbar signifies quantities in the uniform pre-stretched state, and a subscript comma denotes the partial differentiation of the corresponding coordinates. Accordingly, the deformation gradient tensor $\overline{F}_{iA}$ mapping C_0 onto C_1 is given by

$$\overline{F}_{iA} = \frac{\partial \bar{x}_i}{\partial X_A} = \lambda_1 \delta_{i1} \delta_{A1} + \lambda_2 \delta_{i2} \delta_{A2} + \lambda_1^{-1} \lambda_2^{-1} \delta_{i3} \delta_{A3}, \tag{1}$$

where δ_{ij} is the Kronecker delta. The deformation gradient tensor, mapping the reference configuration C_0 to the current configuration C_2, is denoted by F_{iA} and can be expressed as

$$F_{iA} = \frac{\partial x_i}{\partial X_A} = \left(\delta_{ij} + u_{i,j}\right) \overline{F}_{jA}. \tag{2}$$

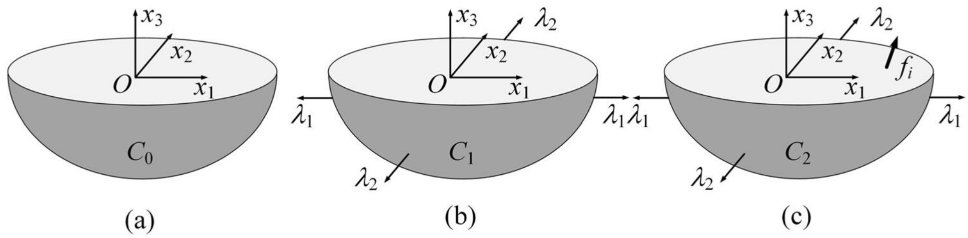

Fig. 1 Schematic illustration of the deformation states of the elastomer for **a** the reference configuration C_0 with no deformation, **b** the intermediate configuration C_1 under biaxial pre-stretch (λ_1, λ_2), and **c** the current configuration C_2, where an incremental surface force f_i is applied to the surface of the pre-stretched elastomer

By applying statistical mechanics to molecular chain networks, the Arruda-Boyce model effectively captures the mechanical response of elastomers across various cross-link densities [32]. Compared to the neo-Hookean model, it effectively captures the strain hardening effect observed under finite deformation [44]. The strain energy density function Ψ for the Arruda-Boyce model is expressed as [32]

$$\Psi = \mu N\left(\frac{\tilde{\lambda}\beta}{\sqrt{N}} + \ln\frac{\beta}{\sinh\beta}\right) - p^*(J-1), \tag{3}$$

with

$$\tilde{\lambda} = \sqrt{\frac{1}{3}I_1},\ \beta = \mathcal{L}^{-1}\left(\sqrt{\frac{I_1}{3N}}\right), \tag{4}$$

where I_1 is the first principal invariant of the left Cauchy-Green strain tensor $B_{ij} = F_{iA}F_{jA}$; J is the determinant of the deformation gradient F_{iA}; the material constants μ and N represent the shear modulus and the number of chain segments, respectively; $\mathcal{L}^{-1}(\cdot)$ is the inverse Langevin function, and the Langevin function is defined as $\mathcal{L}(x) = \coth x - 1/x$. The Lagrange multiplier p^*, associated with the incompressibility constraint $J = 1$, is decomposed as

$$p^* = \overline{p} + \tilde{p}, \tag{5}$$

where $\overline{p}$ denotes the uniform hydrostatic pressure in the intermediate configuration C_1, and $\tilde{p}$ represents the incremental hydrostatic pressure induced by the surface traction in the current configuration C_2. Here, p^* acts as a workless reaction to the kinematic constraint imposed on the deformation field, which can be determined by the equilibrium equations and the boundary conditions. Assuming the incremental displacements are infinitesimal, the displacement gradients $u_{i,j}$, the second-order gradients $u_{i,jk}$, and the incremental pressure $\tilde{p}$ are treated as first-order small quantities of magnitude $O(\kappa)$. In the subsequent analysis, a linearization is performed by neglecting terms of $O(\kappa^2)$ and higher. Consequently, the incompressibility condition in configuration C_2 is reduced to its linearized form as

$$u_{i,i} = 0. \tag{6}$$

Following the strain energy density function defined in Eq. (3), the first Piola–Kirchhoff stress tensor τ_{iA} is derived as

$$\tau_{iA} = \frac{\partial\Psi}{\partial F_{iA}} = \mu\frac{\sqrt{N}\beta}{\sqrt{3I_1}}F_{iA} - p^*F_{Ai}^{-1}. \tag{7}$$

In the absence of body forces, the equilibrium equation $\tau_{iA,A} = 0$ can be reformulated as

$$\left(\tau_{iA}\overline{F}_{jA}\right)_{,j} = 0 \tag{8}$$

When a concentrated force f_i is applied to an area element ds on the surface of the pre-stretched elastomer, the boundary condition is expressed as

$$\tau_{iA}N_A\mathrm{d}S = f_i\delta(x_1)\delta(x_2)\mathrm{d}s \text{ at } x_3 = 0, \tag{9}$$

where $N_A = (0, 0, 1)$ denotes the unit outward normal to the area element dS in the reference configuration C_0, and $\delta(\cdot)$ represents the Dirac delta function used to characterize the concentrated force. By invoking Nanson's formula, $N_A\mathrm{d}S = \overline{F}_{iA}n_i\mathrm{d}s$, Eq. (9) can be transformed into

$$\tau_{iA}\overline{F}_{jA}n_j = f_i\delta(x_1)\delta(x_2) \text{ at } x_3 = 0, \tag{10}$$

where $n_j = (0, 0, 1)$ is the unit outward normal to the area element ds in the intermediate configuration C_1. In C_1, where the surface is free of external traction (i.e., $f_i = 0$), the constant first invariant $\overline{I}_1$ and the corresponding inverse Langevin parameter $\overline{\beta}$ are given by

$$\overline{I}_1 = \lambda_1^2 + \lambda_2^2 + \lambda_1^{-2}\lambda_2^{-2},\ \overline{\beta} = \mathcal{L}^{-1}\left(\sqrt{\frac{\overline{I}_1}{3N}}\right). \tag{11}$$

By substituting Eq. (11) into the boundary condition [Eq. (10)] and considering configuration C_1 (i.e., $f_i = 0$), the constant hydrostatic pressure $\overline{p}$ is readily determined as

$$\overline{p} = \mu \frac{\sqrt{N}\overline{\beta}}{\sqrt{3\overline{I}_1}} \lambda_1^{-2} \lambda_2^{-2}. \tag{12}$$

Next, we consider the perturbed configuration C_2. Based on the definition of the left Cauchy-Green deformation tensor $B_{ij} = F_{iA}F_{jA}$, the first invariant I_1 can be linearized around the pre-stretched state. Retaining only the first-order terms of the displacement gradients, I_1 is expressed as

$$I_1 = B_{ii} = \overline{I}_1 + 2u^*, \tag{13}$$

where u^* represents the linear increment of the strain invariant, defined as

$$u^* = \lambda_1^2 u_{1,1} + \lambda_2^2 u_{2,2} + \lambda_1^{-2}\lambda_2^{-2} u_{3,3}. \tag{14}$$

Based on the definition in Eq. (13) and considering that u^* is a first-order small quantity, the Taylor expansions of $\sqrt{I_1}$ and β are obtained as follows

$$\sqrt{I_1} \approx \sqrt{\overline{I}_1}\left(1 + \frac{1}{\overline{I}_1}u^*\right),\ \beta \approx \overline{\beta} + \tilde{\beta}u^*. \tag{15}$$

where $\tilde{\beta}$ denotes the first-order expansion coefficient of β with respect to u^*. Since the inverse Langevin function lacks a closed-form analytical expression, various approximations have been proposed [45]. In the present study, we adopt the high-accuracy approximation proposed by Kroeger [46], which is given by

$$\mathcal{L}^{-1}(x) = \frac{3x}{\left(1 - x^2\right)\left(1 + \frac{1}{2}x^2\right)} \tag{16}$$

Accordingly, the explicit expressions for $\overline{\beta}$ and $\tilde{\beta}$ can be derived as

$$\overline{\beta} = \frac{3\overline{x}}{\left(1 - \overline{x}^2\right)\left(1 + \frac{1}{2}\overline{x}^2\right)}, \tag{17}$$

$$\tilde{\beta} = \frac{3\overline{x}^4 + \overline{x}^2 + 2}{2N\overline{x}\left(1 - \overline{x}^2\right)^2\left(1 + \frac{1}{2}\overline{x}^2\right)^2}, \tag{18}$$

with the parameter $\overline{x}$ defined as: $\overline{x} = \sqrt{\overline{I}_1/3N}$. Utilizing the approximation $(1+\kappa)^{-1} \approx 1 - \kappa$ since κ is a small quantity, $\tau_{iA}\overline{F}_{jA}$ is expanded as

$$\tau_{iA}\overline{F}_{jA} = \mu \frac{\sqrt{N}\left(\overline{\beta} + \tilde{\beta}u^*\right)}{\sqrt{3\overline{I}_1}\left(1 + u^*/\overline{I}_1\right)} \left(\delta_{ik} + u_{i,k}\right)\overline{F}_{kA}\overline{F}_{jA} - \left(\overline{p} + \tilde{p}\right)\left(\delta_{ki} - u_{k,i}\right)\overline{F}_{Ak}^{-1}\overline{F}_{jA}. \tag{19}$$

Combining Eqs. (8 and 19), we can write the equilibrium equations in a component form as

$$\mu_1\lambda_1^2 u^*_{,1} + \mu_2\left(\lambda_1^2 u_{1,11} + \lambda_2^2 u_{1,22} + \lambda_1^{-2}\lambda_2^{-2}u_{1,33}\right) - \tilde{p}_{,1} = 0, \tag{20}$$

$$\mu_1\lambda_2^2 u^*_{,2} + \mu_2\left(\lambda_1^2 u_{2,11} + \lambda_2^2 u_{2,22} + \lambda_1^{-2}\lambda_2^{-2}u_{2,33}\right) - \tilde{p}_{,2} = 0, \tag{21}$$

$$\mu_1\lambda_1^{-2}\lambda_2^{-2} u^*_{,3} + \mu_2\left(\lambda_1^2 u_{3,11} + \lambda_2^2 u_{3,22} + \lambda_1^{-2}\lambda_2^{-2}u_{3,33}\right) - \tilde{p}_{,3} = 0, \tag{22}$$

with the following definitions

$$\mu_1 = \mu \frac{\sqrt{N}(\sqrt{3\overline{I}_1}\tilde{\beta} - \sqrt{3/\overline{I}_1}\overline{\beta})}{3\overline{I}_1}, \tag{23}$$

$$\mu_2 = \mu \frac{\sqrt{N}\overline{\beta}}{\sqrt{3\overline{I}_1}}. \tag{24}$$

Based on Eqs. (10 and 19), we can further express the boundary conditions in component form

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(u_{1,3} + u_{3,1}\right) = f_1\delta(x_1)\delta(x_2), \tag{25}$$

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(u_{2,3} + u_{3,2}\right) = f_2\delta(x_1)\delta(x_2), \tag{26}$$

$$\mu_1\lambda_1^{-2}\lambda_2^{-2}u^* + 2\mu_2\lambda_1^{-2}\lambda_2^{-2}u_{3,3} - \tilde{p} = f_3\delta(x_1)\delta(x_2). \tag{27}$$

It is noteworthy that in the limit as the strain hardening parameter N approaches infinity, μ_1 approaches 0 and μ_2 approaches μ. In this limiting case, the equilibrium equations Eqs. (20–22) and the corresponding boundary conditions Eqs. (25–27) reduce to those of the neo-Hookean model, which is consistent with the findings reported by He [12]. The governing equations presented above can be solved using the Fourier transform, and the surface Green's function G_{ij} is subsequently obtained. The surface Green's function G_{ij} is defined by the relation $u_i(\mathbf{x}) = G_{ij}(\mathbf{x} - \mathbf{x}')f_j$, where $\mathbf{x} = (x_1, x_2, 0)$ represents a point on the free surface and $\mathbf{x}' = (x_1', x_2', 0)$ denotes the location of the applied concentrated force on the free surface. For the sake of conciseness, the detailed derivation is provided in Appendix A, where we provide the surface Green's function G_{ij} in the frequency domain. Under current conditions, transforming the surface Green's function from the frequency domain to the real domain is non-trivial due to the complexity of the kernels. On the other hand, G_{33} is dominant in the study of contact and adhesion problems. Therefore, a specialized case of equibiaxial pre-stretch is considered, where $\lambda_1 = \lambda_2 = \lambda$, allowing for a closed-form analytical solution. The explicit expression for the surface Green's function component G_{33} for the equibiaxial pre-stretch condition is expressed as

$$G_{33}\left(\mathbf{x}-\mathbf{x}'\right) = \frac{f(\lambda)}{2\pi\mu r}, \tag{28}$$

$$f(\lambda) = \frac{\lambda^4(L+S)}{\frac{\mu_2}{\mu}\left[L^2S^2 + L^2 + S^2 + 3LS + \lambda^6(LS-1) - 2\right] + \frac{\mu_1}{\mu}\left(\lambda^2 - \lambda^{-4}\right)\left(\lambda^6 - 1\right)(LS-1)}, \tag{29}$$

$$r = \sqrt{\left(x_1 - x_1'\right)^2 + \left(x_2 - x_2'\right)^2}, \tag{30}$$

$$L = \sqrt{\frac{\left(\lambda^6+1\right) + \frac{\mu_1}{\mu_2}\left(\lambda^2-\lambda^{-4}\right)\left(\lambda^6-1\right) - \sqrt{\left[\left(\lambda^6+1\right) + \frac{\mu_1}{\mu_2}\left(\lambda^2-\lambda^{-4}\right)\left(\lambda^6-1\right)\right]^2 - 4\lambda^6}}{2}}, \tag{31}$$

$$S = \sqrt{\frac{\left(\lambda^6+1\right) + \frac{\mu_1}{\mu_2}\left(\lambda^2-\lambda^{-4}\right)\left(\lambda^6-1\right) + \sqrt{\left[\left(\lambda^6+1\right) + \frac{\mu_1}{\mu_2}\left(\lambda^2-\lambda^{-4}\right)\left(\lambda^6-1\right)\right]^2 - 4\lambda^6}}{2}}. \tag{32}$$

Equations (28–32) will be used to study the contact problem between a pre-stretched half-space with strain hardening effect and a rigid sphere in Sect. 3.1.

2.2 Surface Deformation in the Thin-Layer Limit

Thin-layer structures with one or both surfaces constrained are widely found in biological tissues and engineered systems [35, 36]. In this section, we derive the surface deformation of an incompressible elastic thin layer with strain hardening effect. Let d_{ref} represent the thickness of the layer before stretching, and d denote the thickness of the layer after stretching. Due to incompressibility, they are related as $d = d_{\text{ref}}/\lambda_1\lambda_2$. Given that the horizontal dimension is assumed to be infinite, we define a characteristic horizontal length ℓ and introduce the dimensionless quantity $\epsilon = d/\ell$. Under the thin-layer assumption $\epsilon \ll 1$, we employ perturbation theory to establish an asymptotic deformation model. The free surface of the thin layer after pre-stretching is located at $x_3 = 0$, while the lower surface is perfectly bonded to a rigid foundation at $x_3 = -d$. The free surface is subjected to prescribed distributed forces $f_i(x_1, x_2)$. While the equilibrium equations remain identical to those derived for the half-space problem [Eqs. (20–22)], the boundary conditions should be reformulated. On the free surface ($x_3 = 0$), the boundary conditions become

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(u_{1,3} + u_{3,1}\right) = f_1\left(x_1, x_2\right), \tag{33}$$

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(u_{2,3} + u_{3,2}\right) = f_2\left(x_1, x_2\right), \tag{34}$$

$$\mu_1\lambda_1^{-2}\lambda_2^{-2}u^* + 2\mu_2\lambda_1^{-2}\lambda_2^{-2}u_{3,3} - \tilde{p} = f_3\left(x_1, x_2\right), \tag{35}$$

and on the bottom surface ($x_3 = -d$), we have

$$u_1 = u_2 = u_3 = 0. \tag{36}$$

The geometric constraints of the thin layer lead to a significant disparity between in-plane and out-of-plane responses. Specifically, both stress and deformation in the lateral directions are expected to be much smaller than their vertical counterparts [47]. Consequently, it is appropriate to assume the following scaling relations [48, 49]: $[u_1], [u_2] \sim \epsilon[u_3]$ and $[f_1(x_1,x_2)], [f_2(x_1,x_2)] \sim \epsilon[f_3(x_1,x_2)]$, where $[\cdot]$ denotes the characteristic magnitude of the respective quantity. Based on this, we introduce the following dimensionless quantities

$$\bar{x}_1 = \frac{x_1}{\ell} = \frac{\epsilon x_1}{d}, \; \bar{x}_2 = \frac{\epsilon x_2}{d}, \; \bar{x}_3 = \frac{x_3}{d}, \tag{37}$$

$$\bar{u}_1 = \frac{u_1}{\epsilon[u_3]}, \; \bar{u}_2 = \frac{u_2}{\epsilon[u_3]}, \; \bar{u}_3 = \frac{u_3}{[u_3]}, \tag{38}$$

$$F_1(\bar{x}_1,\bar{x}_2)=\frac{df_1(\bar{x}_1,\bar{x}_2)}{\epsilon\mu[u_3]},\ F_2(\bar{x}_1,\bar{x}_2)=\frac{df_2(\bar{x}_1,\bar{x}_2)}{\epsilon\mu[u_3]},\ F_3(\bar{x}_1,\bar{x}_2)=\frac{df_3(\bar{x}_1,\bar{x}_2)}{\mu[u_3]}, \tag{39}$$

Based on Eqs. (37–39) and scale analysis of the equilibrium equations [Eqs. (20–22)], the dimensionless quantity associated with $\tilde{p}$ is expressed as

$$P=\frac{d\tilde{p}}{\mu[u_3]}. \tag{40}$$

The dimensionless equilibrium equations can be written as follows

$$\mu_1\lambda_1^2\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_1}{\partial\bar{x}_1^2}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_2}{\partial\bar{x}_2\partial\bar{x}_1}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_3}{\partial\bar{x}_3\partial\bar{x}_1}\right)+\mu_2\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_1}{\partial\bar{x}_1^2}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_1}{\partial\bar{x}_2^2}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_1}{\partial\bar{x}_3^2}\right)-\mu\frac{\partial P}{\partial\bar{x}_1}=0, \tag{41}$$

$$\mu_1\lambda_2^2\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_1}{\partial\bar{x}_1\partial\bar{x}_2}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_2}{\partial\bar{x}_2^2}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_3}{\partial\bar{x}_3\partial\bar{x}_2}\right)+\mu_2\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_2}{\partial\bar{x}_1^2}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_2}{\partial\bar{x}_2^2}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_2}{\partial\bar{x}_3^2}\right)-\mu\frac{\partial P}{\partial\bar{x}_2}=0, \tag{42}$$

$$\mu_1\lambda_1^{-2}\lambda_2^{-2}\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_1}{\partial\bar{x}_1\partial\bar{x}_3}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_2}{\partial\bar{x}_2\partial\bar{x}_3}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_3}{\partial\bar{x}_3^2}\right)+\mu_2\left(\lambda_1^2\frac{\epsilon^2\partial^2\bar{u}_3}{\partial\bar{x}_1^2}+\lambda_2^2\frac{\epsilon^2\partial^2\bar{u}_3}{\partial\bar{x}_2^2}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial^2\bar{u}_3}{\partial\bar{x}_3^2}\right)-\mu\frac{\partial P}{\partial\bar{x}_3}=0. \tag{43}$$

Boundary conditions on the upper surface ($\bar{x}_3=0$) are

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(\frac{\partial\bar{u}_1}{\partial\bar{x}_3}+\frac{\partial\bar{u}_3}{\partial\bar{x}_1}\right)=\mu F_1(\bar{x}_1,\bar{x}_2), \tag{44}$$

$$\mu_2\lambda_1^{-2}\lambda_2^{-2}\left(\frac{\partial\bar{u}_2}{\partial\bar{x}_3}+\frac{\partial\bar{u}_3}{\partial\bar{x}_2}\right)=\mu F_2(\bar{x}_1,\bar{x}_2), \tag{45}$$

$$\mu_1\lambda_1^{-2}\lambda_2^{-2}\left(\lambda_1^2\frac{\epsilon^2\partial\bar{u}_1}{\partial\bar{x}_1}+\lambda_2^2\frac{\epsilon^2\partial\bar{u}_2}{\partial\bar{x}_2}+\lambda_1^{-2}\lambda_2^{-2}\frac{\partial\bar{u}_3}{\partial\bar{x}_3}\right)+2\mu_2\lambda_1^{-2}\lambda_2^{-2}\frac{\partial\bar{u}_3}{\partial\bar{x}_3}-\mu P=\mu F_3(\bar{x}_1,\bar{x}_2), \tag{46}$$

and at the bottom surface ($\bar{x}_3=-1$), we have

$$\bar{u}_1=\bar{u}_2=\bar{u}_3=0. \tag{47}$$

The incompressible condition becomes

$$\frac{\epsilon^2\partial\bar{u}_1}{\partial\bar{x}_1}+\frac{\epsilon^2\partial\bar{u}_2}{\partial\bar{x}_2}+\frac{\partial\bar{u}_3}{\partial\bar{x}_3}=0. \tag{48}$$

Equations (41–48) constitute a complete description of the problem of an incompressible pre-stretched thin layer with strain hardening effect under a distributed force. An inspection of the dimensionless governing equations, boundary conditions, and the incompressibility constraint reveals that the small parameter ϵ appears exclusively in even powers. Accordingly, we seek an approximate solution by proposing an asymptotic expansion of the following form

$$\bar{u}_1=\bar{u}_1^{(0)}+\epsilon^2\bar{u}_1^{(1)}+O(\epsilon^4), \tag{49}$$

$$\bar{u}_2=\bar{u}_2^{(0)}+\epsilon^2\bar{u}_2^{(1)}+O(\epsilon^4), \tag{50}$$

$$\bar{u}_3=\bar{u}_3^{(0)}+\epsilon^2\bar{u}_3^{(1)}+O(\epsilon^4), \tag{51}$$

$$P=P^{(0)}+\epsilon^2P^{(1)}+O(\epsilon^4). \tag{52}$$

We solve the above equations order by order. At leading order, the solution to Eqs. (41–48) is

$$\bar{u}_1^{(0)}=\frac{\mu}{\mu_2}\lambda_1^2\lambda_2^2\left\{-\frac{\partial F_3(\bar{x}_1,\bar{x}_2)}{2\partial\bar{x}_1}\bar{x}_3^2+F_1(\bar{x}_1,\bar{x}_2)\bar{x}_3+\frac{\partial F_3(\bar{x}_1,\bar{x}_2)}{2\partial\bar{x}_1}+F_1(\bar{x}_1,\bar{x}_2)\right\}, \tag{53}$$

$$\bar{u}_2^{(0)}=\frac{\mu}{\mu_2}\lambda_1^2\lambda_2^2\left\{-\frac{\partial F_3(\bar{x}_1,\bar{x}_2)}{2\partial\bar{x}_2}\bar{x}_3^2+F_2(\bar{x}_1,\bar{x}_2)\bar{x}_3+\frac{\partial F_3(\bar{x}_1,\bar{x}_2)}{2\partial\bar{x}_2}+F_2(\bar{x}_1,\bar{x}_2)\right\}, \tag{54}$$

$$\bar{u}_3^{(0)}=0, \tag{55}$$

$$P^{(0)} = -F_3(\bar{x}_1, \bar{x}_2). \tag{56}$$

The normal displacement at leading order $\overline{u}_3^{(0)}$ is 0, which is due to the incompressibility of the material, consistent with the discussion of Chandler and Vella on Poisson's ratio of thin layers [48]. At the next order, the incompressibility condition becomes

$$\frac{\partial \overline{u}_1^{(0)}}{\partial \bar{x}_1} + \frac{\partial \overline{u}_2^{(0)}}{\partial \bar{x}_2} + \frac{\partial \overline{u}_3^{(1)}}{\partial \bar{x}_3} = 0. \tag{57}$$

Combining Eq. (57) and boundary conditions $\overline{u}_3^{(1)} = 0$ at $\bar{x}_3 = -1$, we can obtain the normal displacement of the upper surface at the next order as

$$\overline{u}_3^{(1)}(\bar{x}_1, \bar{x}_2, 0) = -\frac{\mu}{\mu_2}\lambda_1^2\lambda_2^2 \left\{ \frac{1}{3}\left[\frac{\partial^2 F_3(\bar{x}_1,\bar{x}_2)}{\partial \bar{x}_1^2} + \frac{\partial^2 F_3(\bar{x}_1,\bar{x}_2)}{\partial \bar{x}_2^2}\right] + \frac{1}{2}\left[\frac{\partial F_1(\bar{x}_1,\bar{x}_2)}{\partial \bar{x}_1} + \frac{\partial F_2(\bar{x}_1,\bar{x}_2)}{\partial \bar{x}_2}\right]\right\}. \tag{58}$$

Further, we can write the displacement of the upper surface in dimensional terms as

$$u_3(x_1, x_2, 0) = -\frac{\lambda_1^2\lambda_2^2}{\mu_2} \left\{ \frac{d^3}{3}\left[\frac{\partial^2 f_3(x_1,x_2)}{\partial x_1 \partial x_1} + \frac{\partial^2 f_3(x_1,x_2)}{\partial x_2 \partial x_2}\right] + \frac{d^2}{2}\left[\frac{\partial f_1(x_1,x_2)}{\partial x_1} + \frac{\partial f_2(x_1,x_2)}{\partial x_2}\right]\right\}. \tag{59}$$

Equation (59) will be used to investigate the contact problem between a pre-stretched thin layer bonded to a rigid substrate with strain hardening effect and a rigid sphere in Sect. 3.2.

3 Examples and Analysis

3.1 Indentation of a Half-Space

We first apply the surface Green's function G_{33} [Eqs. (28–32)] for the half-space limit to an indentation problem. In a Cartesian coordinate system $Ox_1x_2x_3$, the elastomer occupies an infinite half-space defined by $x_3 \le 0$. The material behavior is described by an incompressible Arruda-Boyce hyperelastic model. The elastomer is pre-stretched along the x_1- and x_2 -directions with equal principal stretch λ, while a rigid sphere makes contact with the elastomer. The radius of the sphere is R. Friction and adhesion between the two surfaces are neglected. This assumption allows us to focus on the influence of pre-stretch and strain hardening on the indentation response. It should be noted that friction may introduce tangential tractions, partial slip, and normal-tangential coupling, while adhesion may modify the contact radius and pressure distribution, thereby influencing the indentation response. According to Hertzian contact theory, we assume the pressure distribution p takes the form

$$p(\mathbf{x}) = p_0\left(1 - r^2/a^2\right)^{1/2}, \tag{60}$$

where $r = |\mathbf{x}|$, a is the radius of the contact region, and p_0 denotes the positive pressure constant. The pressure distribution immediately gives the total force acting on the rigid sphere as: $F = 2\pi p_0 a^2/3$. The normal surface displacement $u_3(\mathbf{x})$ of the elastomer surface induced by contact can be obtained by the surface integral of the product $-G_{33}(\mathbf{x} - \mathbf{x}')p(\mathbf{x}')$ over the contact region as

$$u_3(\mathbf{x}) = -\frac{\pi f(\lambda) p_0}{8\mu a}(2a^2 - r^2). \tag{61}$$

For a spherical indenter with radius R and indentation depth δ, the displacement on the surface of the elastomer inside the contact region induced by the indenter has the form

$$u_3(\mathbf{x}) = -\delta + \frac{r^2}{2R}. \tag{62}$$

Comparing Eqs. (61 and 62), we can conclude that

$$p_0 = \frac{4\mu a}{\pi R f(\lambda)},\ \delta = \frac{\pi a f(\lambda) p_0}{4\mu}. \tag{63}$$

With Eq. (63), we can obtain the contact force and displacement of the rigid sphere as

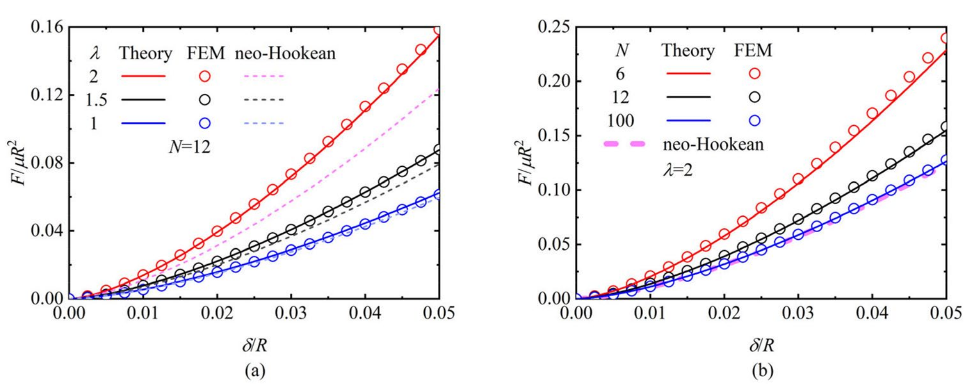

Fig. 2 Normalized indentation force $F/\mu R^2$ as a function of normalized indentation depth δ/R for **a** different pre-stretch λ and **b** different strain hardening parameter N. The solid lines and circles represent the analytical solutions and FEM results, respectively. The results for the neo-Hookean model (dashed lines) are adapted from He and Ding [18]

$$F = \frac{8\mu a^3}{3Rf(\lambda)},\ \delta = \frac{a^2}{R}. \tag{64}$$

To verify the analytical model mentioned above, we construct axisymmetric finite element simulations using the commercial software ABAQUS. The incompressible Arruda-Boyce model is implemented with a user-defined subroutine (UMAT) rather than adopting the built-in model to be consistent with the inverse Langevin function [Eq. (16)]. Prior to contact, the elastomer undergoes pre-stretch, which introduces both pre-strain and pre-stress in the elastomer. The dimensions of the pre-stretched elastomer are set to $50R \times 50R$ (where R is the radius of the rigid sphere) to approximate a half-space. The substrate is discretized into 37,330 four-node bilinear axisymmetric hybrid elements (CAX4H), with the mesh locally refined to $0.003R$ near the contact region to ensure numerical convergence. Normal displacement constraints are set on the side and bottom of the elastomer. A comparison of the results between the finite element model and the analytical model is shown in Fig. 2.

Figure 2a presents the normalized indentation force as a function of the normalized displacement for different pre-stretches ($\lambda = 1, 1.5, 2$). The dashed curves correspond to the predictions of the neo-Hookean constitutive model (i.e., without strain hardening) [18]. In the absence of pre-stretch ($\lambda = 1$), the two constitutive descriptions yield nearly indistinguishable responses. As the pre-stretch increases, the discrepancy between the strain-hardened model and the neo-Hookean model becomes progressively more pronounced. For a given indentation depth, the strain-hardened elastomer consistently exhibits a higher indentation force than the neo-Hookean counterpart.

With the pre-stretch fixed at $\lambda = 2$, Fig. 2b further compares the normalized force–displacement response for different values of the strain-hardening parameter N. At the same indentation depth, a smaller N leads to a larger indentation force, reflecting stronger strain hardening. In the limit of large N (e.g., $N = 100$), the present model asymptotically recovers the neo-Hookean response (dashed curves) [18]. The theoretical predictions agree well with finite element simulations.

3.2 Indentation of a Thin Layer

In this section, we investigate the indentation of a biaxially stretched thin elastic layer with strain hardening effect by a rigid spherical indenter of radius R. The layer is perfectly bonded to a rigid substrate at the lower surface, whereas the upper surface is in frictionless and non-adhesive contact with the rigid sphere. According to Eq. (59), under an arbitrary biaxial pre-stretch, the surface displacement response of the thin layer remains isotropic up to $O(\epsilon^2)$. The surface deformation induced by the prescribed traction is governed by the product of the principal stretches $\lambda_1\lambda_2$ and the strain hardening modulus μ_2. Let δ and a denote the indentation depth and the contact radius, respectively. Under the thin-layer assumption ($R \gg a \gg d$), the surface deformation [Eq. (59)] can be formulated using the Laplacian operator ∇^2 as follows

$$u_3(\mathbf{x}) = -\frac{\lambda_1^2\lambda_2^2}{\mu_2}\frac{d^3}{3}\nabla^2 p = -\frac{1}{\lambda_1\lambda_2\mu_2}\frac{d_{\mathrm{ref}}^3}{3}\nabla^2 p, \tag{65}$$

where p denotes the normal pressure, defined as positive in the downward direction such that $p = -f_3$. Within the contact region ($r \le a$), the vertical surface displacement $u_3(\mathbf{x})$ is governed by the geometry of the spherical indenter

$$u_3(\mathbf{x}) = -\delta + \frac{r^2}{2R}. \tag{66}$$

The boundary conditions at the edge of the contact region are given by [50, 51]

$$p(a) = 0, p'(a) = 0. \tag{67}$$

Combining Eqs. (65–67), the closed-form solution for the pressure distribution p is obtained as

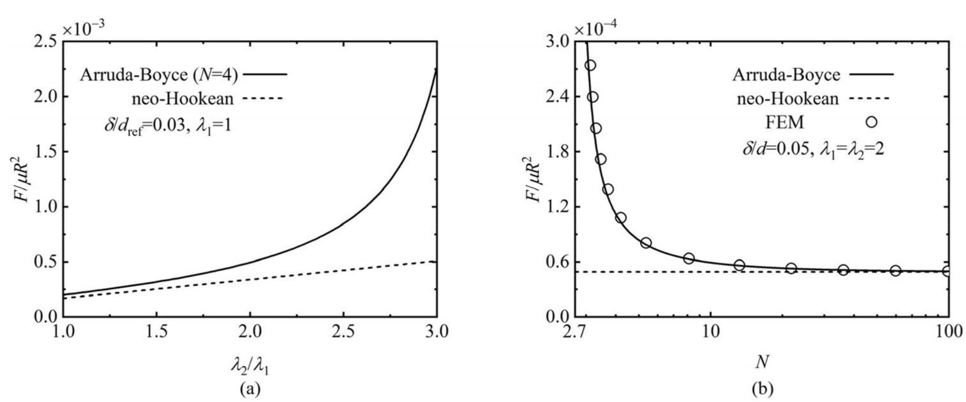

Fig. 3 Normalized indentation force $F/\mu R^2$ as a function of **a** pre-stretch ratio λ_2/λ_1 and **b** strain hardening parameter N. Solid curves denote the Arruda-Boyce model, while dashed curves represent the neo-Hookean model ($N \to \infty$, $\mu_2 \to \mu$), and circles in **b** indicate the finite element simulation results

$$p(\mathbf{x}) = \frac{3\lambda_1\lambda_2\mu_2}{d_{\text{ref}}^3}\left(\frac{\delta}{4}\left(a^2 - r^2\right) + \frac{1}{32R}\left(r^4 - a^4\right)\right), \tag{68}$$

where the indentation depth δ is related to the contact radius a through the relationship: $\delta = a^2/4R$. The total indentation force F acting on the rigid spherical indenter is determined by integrating the contact pressure over the contact region ($r \leq a$) as

$$F = \int_0^{2\pi}\int_0^a pr\mathrm{d}r\mathrm{d}\theta = \frac{2\pi\mu_2 R^2\delta^3}{\lambda_1^2\lambda_2^2 d^3} = \frac{2\pi\lambda_1\lambda_2\mu_2 R^2\delta^3}{d_{\text{ref}}^3}. \tag{69}$$

In the absence of pre-stretch ($\lambda_1 = \lambda_2 = 1$) and in the limit where $N \to \infty$, Eq. (69) recovers the solutions for the indentation of an incompressible thin layer bonded to a rigid substrate as established by Yang [50] and Chadwick [51].

Figure 3a illustrates the normalized indentation force as a function of the pre-stretch state for thin elastic layers with and without strain hardening. The first principal stretch is fixed at $\lambda_1 = 1$, and a constant dimensionless indentation depth of $\delta/d_{\text{ref}} = 0.03$. The results demonstrate that for the neo-Hookean model, the normalized indentation force is linearly proportional to the product of the principal stretches $\lambda_1\lambda_2$. In contrast, when strain hardening effect is incorporated, the normalized force exhibits a pronounced non-linear escalation with increasing stretch ratio. The normalized indentation force is related to the product of the principal stretching $\lambda_1\lambda_2$ and the strain hardening modulus μ_2, where μ_2 is a monotonically increasing function of $\bar{x}$ [according to Eqs. (17 and 24), $\mu_2 = \frac{\mu}{(1-\bar{x}^2)(1+0.5\bar{x}^2)}$], and $\bar{x}$ is larger in the stretched state than in the undeformed state, which means that μ_2 is larger after stretching the thin layer. The growth of normalized indentation force becomes particularly rapid as the stretch ratio becomes large, where the finite extensibility of the polymer chains leads to a sharp increase in the effective stiffness of the elastomer.

Figure 3b presents the relationship between the normalized indentation force and the strain hardening parameter N for a fixed biaxial pre-stretch ($\lambda_1 = \lambda_2 = 2$). When N approaches 2.7, the normalized indentation force tends to infinity. This singular behavior arises because the corresponding chain-stretch measure $\bar{x}$ approaches 1. When N tends to infinity, the normalized indentation force converges to that of the neo-Hookean model. To verify the analytical model, an axisymmetric finite element model is constructed following the procedure described in Sect. 3.1. The ratio of the initial thickness of the elastomer, its width, and the radius of the sphere is set to 1:100:8000, so that the thin-layer assumption is satisfied. The elastomer is discretized using a uniform mesh of four-node bilinear axisymmetric hybrid elements (CAX4H). For the present problem, the finite element simulation results are sensitive to the number of elements through the thickness but insensitive to that in the lateral direction. In the simulations, 10,000 elements are used in the lateral direction. A mesh convergence study is conducted by progressively increasing the number of elements through the thickness, and the numerical results are found to converge when no fewer than 120 elements are used. Accordingly, 1.2×10^6 elements are adopted in the present simulations, providing satisfactory convergence. The finite element simulation results are shown as circles in Fig. 3b, which agree with the theoretical predictions.

4 Conclusion

In summary, this study investigates the incremental elasticity of pre-stretched, strain hardening elastomers characterized by the incompressible Arruda-Boyce hyperelastic model. By employing the Fourier transform and perturbation analysis, we derive the surface Green's function and surface deformation for two limiting cases: a half-space elastomer and a thin layer bonded to a rigid substrate. To illustrate the theoretical framework, spherical indentation problems are analyzed for both configurations. We obtained analytical expressions for the relationship between indentation force and indentation depth under the two limiting conditions. In both cases, the indentation force–displacement relationship is jointly influenced by the pre-stretch state and the number of chain segments, i.e., the hardening parameter N. A larger pre-stretch state makes it easier to approach the limit of molecular chain stretching, resulting in a stiffer mechanical behavior of the material, which in turn affects the indentation behavior. However, if the number of chain segments is very large, a smaller pre-stretch cannot allow the molecular chains to fully extend; therefore, the indentation behavior of this material asymptotically follows that of the neo-Hookean constitutive model. The fundamental surface formulas derived in this paper provide an effective tool for characterizing the mechanical properties of strain hardening soft materials and offer a theoretical basis for investigating other phenomena of pre-stretched soft materials with strain hardening effect, such as adhesion, lubrication, and wetting.

Appendix A

Equations (20–22 and 25–27) formulate the boundary value problem of a concentrated force applied to the free surface of an incompressible pre-stretched half-space with strain hardening effect. In this section, we employ Fourier transform to derive the surface Green's function. Because of incompressibility, we introduce two stream functions $\phi_1 = \phi_1(x_i)$ and $\phi_2 = \phi_2(x_i)$, which provide a divergence-free representation of the displacement field. Accordingly, the displacement components u_i can be expressed in terms of ϕ_1 and ϕ_2 as

$$u_1 = \phi_{1,3},\ u_2 = \phi_{2,3},\ u_3 = -\phi_{1,1} - \phi_{2,2}. \tag{A.1}$$

The functions ϕ_1, ϕ_2, and $\tilde{p}$ vanish at infinity. Their Fourier transforms are given by

$$\Phi_1(\xi_1, \xi_2, x_3) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \phi_1(x_1, x_2, x_3)e^{i\boldsymbol{\xi}\cdot\mathbf{x}}dx_1dx_2, \tag{A.2}$$

$$\Phi_2(\xi_1, \xi_2, x_3) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \phi_2(x_1, x_2, x_3)e^{i\boldsymbol{\xi}\cdot\mathbf{x}}dx_1dx_2, \tag{A.3}$$

$$P(\xi_1, \xi_2, x_3) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \tilde{p}(x_1, x_2, x_3)e^{i\boldsymbol{\xi}\cdot\mathbf{x}}dx_1dx_2, \tag{A.4}$$

where $i = \sqrt{-1}$ and $\boldsymbol{\xi}$ is the wave-vector with components ξ_1 and ξ_2. Therefore, the equilibrium equations and boundary conditions in the frequency domain are expressed as follows

$$-\mu_1\lambda_1^4\lambda_2^2\left(\gamma_1\xi_1^2\Phi_{1,3} + \gamma_2\xi_1\xi_2\Phi_{2,3}\right) + \mu_2\Phi_{1,333} - \mu_2\eta^2\Phi_{1,3} + i\lambda_1^2\lambda_2^2\xi_1 P = 0, \tag{A.5}$$

$$-\mu_1\lambda_1^2\lambda_2^4\left(\gamma_1\xi_1\xi_2\Phi_{1,3} + \gamma_2\xi_2^2\Phi_{2,3}\right) + \mu_2\Phi_{2,333} - \mu_2\eta^2\Phi_{2,3} + i\lambda_1^2\lambda_2^2\xi_2 P = 0, \tag{A.6}$$

$$-\mu_1\left(\gamma_1\xi_1\Phi_{1,33} + \gamma_2\xi_2\Phi_{2,33}\right) + \mu_2\left(\xi_1\Phi_{1,33} + \xi_2\Phi_{2,33}\right) - \mu_2\eta^2\left(\xi_1\Phi_1 + \xi_2\Phi_2\right) + i\lambda_1^2\lambda_2^2 P_{,3} = 0, \tag{A.7}$$

and

$$\mu_2\Phi_{1,33} + \mu_2\xi_1\left(\xi_1\Phi_1 + \xi_2\Phi_2\right) = \lambda_1^2\lambda_2^2 f_1, \tag{A.8}$$

$$\mu_2\Phi_{2,33} + \mu_2\xi_2\left(\xi_1\Phi_1 + \xi_2\Phi_2\right) = \lambda_1^2\lambda_2^2 f_2, \tag{A.9}$$

$$i\mu_1\left(\gamma_1\xi_1\Phi_{1,3} + \gamma_2\xi_2\Phi_{2,3}\right) + 2i\mu_2\left(\xi_1\Phi_{1,3} + \xi_2\Phi_{2,3}\right) - \lambda_1^2\lambda_2^2 P = \lambda_1^2\lambda_2^2 f_3, \tag{A.10}$$

with

$$\gamma_1 = \lambda_1^2 - \lambda_1^{-2}\lambda_2^{-2},\ \gamma_2 = \lambda_2^2 - \lambda_1^{-2}\lambda_2^{-2},\ \eta = \lambda_1\lambda_2\sqrt{\lambda_1^2\xi_1^2 + \lambda_2^2\xi_2^2}. \tag{A.11}$$

After some algebra, the general solution of the equilibrium equations [Eqs. (A.5–A.7)] can be obtained as

$$\Phi_1 = -\gamma_2\xi_2 c_1 e^{\eta x_3} + A_1 c_2 e^{l x_3} + A_2 c_3 e^{s x_3}, \tag{A.12}$$

$$\Phi_2 = \gamma_1\xi_1 c_1 e^{\eta x_3} + B_1 c_2 e^{l x_3} + B_2 c_3 e^{s x_3}, \tag{A.13}$$

$$P = i\mu_2 l D_1 c_2 e^{l x_3} + i\mu_2 s D_2 c_3 e^{s x_3}, \tag{A.14}$$

where c_i represent the coefficients to be determined from the boundary conditions [Eqs. (A.8–A.10)]. The expressions for A_1, A_2, B_1, B_2, D_1, and D_2 are as follows

$$A_1 = \xi_1\left(\mu_1\lambda_1^2\lambda_2^2\gamma_2\xi_2^2\left(\lambda_1^2 - \lambda_2^2\right) + \mu_2\left(l^2 - \eta^2\right)\right)\lambda_1^2\lambda_2^2, \tag{A.15}$$

$$A_2 = \xi_1\left(\mu_1\lambda_1^2\lambda_2^2\gamma_2\xi_2^2\left(\lambda_1^2 - \lambda_2^2\right) + \mu_2\left(s^2 - \eta^2\right)\right)\lambda_1^2\lambda_2^2, \tag{A.16}$$

$$B_1 = \xi_2\left(\mu_1\lambda_1^2\lambda_2^2\gamma_1\xi_1^2\left(\lambda_2^2 - \lambda_1^2\right) + \mu_2\left(l^2 - \eta^2\right)\right)\lambda_1^2\lambda_2^2, \tag{A.17}$$

$$B_2 = \xi_2\left(\mu_1\lambda_1^2\lambda_2^2\gamma_1\xi_1^2\left(\lambda_2^2 - \lambda_1^2\right) + \mu_2\left(s^2 - \eta^2\right)\right)\lambda_1^2\lambda_2^2, \tag{A.18}$$

$$D_1 = \left(l^2 - \eta^2\right)\left[-\mu_1\lambda_1^2\lambda_2^2\left(\gamma_1\lambda_1^2\xi_1^2 + \gamma_2\lambda_2^2\xi_2^2\right) + \mu_2\left(l^2 - \eta^2\right)\right], \tag{A.19}$$

$$D_2 = \left(s^2 - \eta^2\right)\left[-\mu_1\lambda_1^2\lambda_2^2\left(\gamma_1\lambda_1^2\xi_1^2 + \gamma_2\lambda_2^2\xi_2^2\right) + \mu_2\left(s^2 - \eta^2\right)\right]. \tag{A.20}$$

The explicit expressions for l and s are given by

$$l = \sqrt{\frac{\Omega - \sqrt{\Omega^2 - 4\mu_2\left[\mu_1\lambda_1^2\lambda_2^2\xi_1^2\xi_2^2\left(\lambda_1^2 - \lambda_2^2\right)^2 + \mu_2\xi^2\eta^2\right]}}{2\mu_2}}, \tag{A.21}$$

$$s = \sqrt{\frac{\Omega + \sqrt{\Omega^2 - 4\mu_2\left[\mu_1\lambda_1^2\lambda_2^2\xi_1^2\xi_2^2\left(\lambda_1^2 - \lambda_2^2\right)^2 + \mu_2\xi^2\eta^2\right]}}{2\mu_2}}, \tag{A.22}$$

with

$$\Omega = \mu_2\left(\eta^2 + \xi^2\right) + \mu_1\lambda_1^2\lambda_2^2\left(\gamma_1\lambda_1^2\xi_1^2 + \gamma_2\lambda_2^2\xi_2^2\right) - \mu_1\left(\gamma_1\xi_1^2 + \gamma_2\xi_2^2\right), \tag{A.23}$$

$$\xi = \sqrt{\xi_1^2 + \xi_2^2}. \tag{A.24}$$

Substituting the general solution [Eqs. (A.12–A.14)] into the boundary conditions [Eqs. (A.8–A.10)], we obtain the following algebraic expressions

$$d_{11}c_1 + d_{12}c_2 + d_{13}c_3 = \frac{\lambda_1^2\lambda_2^2 f_1}{\mu_2}, \tag{A.25}$$

$$d_{21}c_1 + d_{22}c_2 + d_{23}c_3 = \frac{\lambda_1^2\lambda_2^2 f_2}{\mu_2}, \tag{A.26}$$

$$d_{31}c_1 + d_{32}c_2 + d_{33}c_3 = \frac{\mathrm{i}\lambda_1^2\lambda_2^2 f_3}{\mu_2}. \tag{A.27}$$

with

$$d_{11} = \xi_2\left[\left(\gamma_1 - \gamma_2\right)\xi_1^2 - \gamma_2\eta^2\right], \tag{A.28}$$

$$d_{21} = \xi_1\left[\left(\gamma_1 - \gamma_2\right)\xi_2^2 + \gamma_1\eta^2\right], \tag{A.31}$$

$$d_{22} = \left(l^2 + \xi_2^2\right)B_1 + \xi_1\xi_2 A_1, \tag{A.32}$$

$$d_{23} = \left(s^2 + \xi_2^2\right)B_2 + \xi_1\xi_2 A_2, \tag{A.33}$$

$$d_{31} = 2\left(\gamma_2 - \gamma_1\right)\xi_1\xi_2\eta, \tag{A.34}$$

$$d_{32} = \left[\left(\frac{\mu_1}{\mu_2}\gamma_1 - 2\right)\xi_1 A_1 + \left(\frac{\mu_1}{\mu_2}\gamma_2 - 2\right)\xi_2 B_1 + \lambda_1^2\lambda_2^2 D_1\right] l, \tag{A.35}$$

$$d_{33} = \left[\left(\frac{\mu_1}{\mu_2}\gamma_1 - 2\right)\xi_1 A_2 + \left(\frac{\mu_1}{\mu_2}\gamma_2 - 2\right)\xi_2 B_2 + \lambda_1^2\lambda_2^2 D_2\right] s. \tag{A.36}$$

With the above equations [Eqs. (A.25–A.36)], we can solve for $c_i (i = 1, 2, 3)$ using Cramer's rule $c_i = \frac{\Delta_i}{\Delta}$, where

$$\Delta = \begin{vmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{vmatrix}, \ \Delta_1 = \begin{vmatrix} \frac{\lambda_1^2\lambda_2^2 f_1}{\mu_2} & d_{12} & d_{13} \\ \frac{\lambda_1^2\lambda_2^2 f_2}{\mu_2} & d_{22} & d_{23} \\ \frac{\mathrm{i}\lambda_1^2\lambda_2^2 f_3}{\mu_2} & d_{32} & d_{33} \end{vmatrix}, \ \Delta_2 = \begin{vmatrix} d_{11} & \frac{\lambda_1^2\lambda_2^2 f_1}{\mu_2} & d_{13} \\ d_{21} & \frac{\lambda_1^2\lambda_2^2 f_2}{\mu_2} & d_{23} \\ d_{31} & \frac{\mathrm{i}\lambda_1^2\lambda_2^2 f_3}{\mu_2} & d_{33} \end{vmatrix}, \ \Delta_3 = \begin{vmatrix} d_{11} & d_{12} & \frac{\lambda_1^2\lambda_2^2 f_1}{\mu_2} \\ d_{21} & d_{22} & \frac{\lambda_1^2\lambda_2^2 f_2}{\mu_2} \\ d_{31} & d_{32} & \frac{\mathrm{i}\lambda_1^2\lambda_2^2 f_3}{\mu_2} \end{vmatrix}. \tag{A.37}$$

$$d_{12} = \left(l^2 + \xi_1^2\right)A_1 + \xi_1\xi_2 B_1, \tag{A.29}$$

$$d_{13} = \left(s^2 + \xi_1^2\right)A_2 + \xi_1\xi_2 B_2, \tag{A.30}$$

Once we have solved for c_i, the boundary value problem of a concentrated force f_i acting on the surface of the pre-stretched half-space with strain hardening effect is complete. The nine components of the surface Green's function G_{ij} in real space can be obtained by evaluating the following integrals

$$G_{11}\left(\mathbf{x} - \mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\Phi_{1,3}\right)|_{x_3=0, f_1=1, f_2=f_3=0}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot(\mathbf{x}-\mathbf{x}')}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.38}$$

$$G_{22}\left(\mathbf{x} - \mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\Phi_{2,3}\right)|_{x_3=0, f_2=1, f_1=f_3=0}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot(\mathbf{x}-\mathbf{x}')}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.39}$$

$$G_{33}\left(\mathbf{x} - \mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\mathrm{i}\xi_1\Phi_1 + \mathrm{i}\xi_2\Phi_2\right)|_{x_3=0, f_3=1, f_1=f_2=0}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot(\mathbf{x}-\mathbf{x}')}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.40}$$

$$G_{12}\left(\mathbf{x} - \mathbf{x}'\right) = G_{21}\left(\mathbf{x} - \mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\Phi_{1,3}\right)|_{x_3=0, f_2=1, f_1=f_3=0}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot(\mathbf{x}-\mathbf{x}')}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.41}$$

$$G_{23}\left(\mathbf{x}-\mathbf{x}'\right) = -G_{32}\left(\mathbf{x}-\mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\Phi_{2,3}\right)|_{x_3=0,f_3=1,f_1=f_2=0}\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot\left(\mathbf{x}-\mathbf{x}'\right)}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.42}$$

$$G_{13}\left(\mathbf{x}-\mathbf{x}'\right) = -G_{31}\left(\mathbf{x}-\mathbf{x}'\right) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(\Phi_{1,3}\right)|_{x_3=0,f_3=1,f_1=f_2=0}\mathrm{e}^{-\mathrm{i}\boldsymbol{\xi}\cdot\left(\mathbf{x}-\mathbf{x}'\right)}\mathrm{d}\xi_1\mathrm{d}\xi_2, \tag{A.43}$$

where the integration kernels in Eqs. (A.38–A.43) take the form of the surface Green's function in the transformed space.

Acknowledgements This work is supported by the Fundamental Research Funds for the Central Universities (Grant No. 226-2025-00137), the National Natural Science Foundation of China (Grant No. 12321002) and the 111 Project, China (Grant No. B21034).

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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